

SentimentLens: Reconciling Sentiment and Ratings via Dual-Modality in the Hospitality Sector

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Abstract—Online travel platforms generate vast volumes of user-generated hotel reviews, offering rich opportunities to understand traveler experiences at scale. However, transforming unstructured textual feedback into structured, actionable insights remains a challenging task. This paper presents *SentimentLens*, a scalable analysis system based on Aspect-Based Sentiment Analysis that performs knowledge extraction from unstructured hotel reviews and organizes them into interpretable service categories. *SentimentLens* integrates aspect term extraction, aspect sentiment classification, semantic category assignment, and multi-level analytical modules to support region-level, hotel-level, and category-level evaluation. The system is designed to operate across different geographic contexts and hospitality settings. To demonstrate its practical utility, we apply *SentimentLens* to a large real-world dataset of over 10,000 publicly available hotel reviews. Through extensive analysis, the framework reveals how traveler sentiment varies across regions, service categories, and hotel archetypes. We further implement a cross-modal reconciliation of textual sentiment and numerical ratings to identify latent operational conflicts, structural inconsistencies in service quality, and high-impact improvement opportunities using importance–performance and entropy-based analyses. The results show that *SentimentLens* effectively transforms large-scale unstructured reviews into actionable intelligence, supporting data-driven decision-making for hospitality management and tourism policy. While demonstrated using a national case study, the proposed system is generalizable to other destinations and review-driven service domains.

Index Terms—Information Fusion, Information Integration, Aspect-Based Sentiment Analysis, Decision Support Systems, Knowledge Discovery

I. INTRODUCTION

Online travel platforms produce large volumes of user-generated hotel feedback in two complementary forms: (i) unstructured textual reviews describing guest experiences in detail, and (ii) structured metadata such as star ratings and trip types. Star ratings provide a quick summary of overall satisfaction but often hide the specific reasons behind positive or negative experiences, while textual reviews contain rich explanations yet are difficult to analyze systematically at scale. Therefore, an effective tourism analytics system should jointly leverage both modalities to produce reliable, interpretable, and actionable insights [1]. Aspect-Based Sentiment Analysis (ABSA) [2], [3] is a widely adopted approach for extracting fine-grained opinions from review text by identifying specific aspects and predicting the sentiment expressed toward each one [4], [5]. However, many ABSA-driven studies stop at

producing predictions, without integrating them into a broader workflow that supports region-level comparison, hotel benchmarking, and decision-making. Moreover, relying solely on either modality risks missing patterns that only become visible when both signals are combined.

In this work, we present *SentimentLens*, a scalable information integration framework that performs multimodal fusion across two heterogeneous data streams: (i) a text-based pipeline that applies ABSA to extract aspect-level sentiment and organizes aspects into standardized service categories, and (ii) a rating-based pipeline that analyzes star rating distributions and traveler trip types with province-wise comparisons and statistical significance testing. A fusion layer then reconciles insights across both modalities, highlighting agreements and discrepancies. To demonstrate its practical utility, we conduct a large-scale case study using more than 10,000 publicly available hotel reviews covering hotels across the nine provinces of Sri Lanka, illustrating how *SentimentLens* uncovers province-level satisfaction differences, category-level strengths and weaknesses, hotel archetypes, and targeted improvement opportunities.

The main contributions of this paper are:

- We propose *SentimentLens*, a unified cross-modal reconciliation framework that integrates hotel review text and structured ratings to extract actionable insights at scale.
- We introduce a systematic approach for aligning aspect-level sentiment with numerical rating signals, enabling the identification of inconsistencies, latent service gaps, and hidden performance issues across modalities.
- We develop an end-to-end analytical pipeline that combines ABSA-based aspect extraction, semantic category alignment, and multi-level aggregation across provinces, hotels, and service categories for interpretable analysis.
- We validate the proposed framework through a large-scale real-world case study, demonstrating that cross-modal integration improves interpretability and reveals insights that are not observable from single-modality analysis alone.

While this paper focuses on hotels, the proposed system generalizes to other review-driven service domains (e.g., restaurants, attractions, airlines) where both text feedback and numerical ratings are available.

II. LITERATURE REVIEW

Prior research consistently identifies a common set of service dimensions that drive customer satisfaction in the hotel industry including room quality staff and reception services food and beverage offerings location convenience and value for money [6], [7]. While these attributes significantly influence satisfaction and loyalty empirical studies show that their performance is often uneven with critical factors such as cleanliness reliability and restaurant services frequently underperforming despite their high importance [7]. This highlights persistent structural gaps in hotel service delivery and the need for targeted improvement strategies.

Recent advances in automated text analysis have enabled large-scale exploitation of online hotel reviews. Studies show that negative reviews concentrate on a few key issues such as cleanliness, staff behavior, and pricing, while positive reviews span a wider range of experiences. Moreover, negative sentiment has a disproportionately strong impact on perceived service quality and behavioral intentions [8]. To capture such fine-grained insights, aspect-based sentiment analysis (ABSA) has emerged as a dominant approach, linking sentiment polarity to specific hotel attributes such as service, food, cleanliness, and location [9]. Although recent work highlights rapid progress in transformer-based ABSA and improved handling of implicit aspects [10], limitations remain, including fragmented pipelines, weak integration with rating metadata, and limited focus on managerial decision support.

Deep learning approaches now dominate ABSA in the hospitality domain. Models such as Convolutional Neural Networks and Bidirectional Long Short-Term Memory networks outperform traditional methods for aspect extraction and sentiment classification [9]. More recent studies leverage transformer embeddings with clustering and semantic similarity to improve implicit aspect detection and contextual understanding [10]. Emerging work also integrates ABSA into decision support systems, including dashboards, recommender systems, and multi-criteria models, to assist managers in identifying strengths, weaknesses, and investment priorities.

However, these decision-support applications remain limited in scope. Most ABSA studies still focus on isolated modeling tasks or single-platform analyses and rarely perform systematic regional or cross-hotel aggregation, which restricts their practical utility for hotel chains and destination-level strategy development. Although recent advances in location-specific modeling, hotel similarity clustering via aspect-aware embeddings, and cross-language/multi-agent frameworks have begun to address these limitations by uncovering hidden service patterns across geographies and languages, unified end-to-end analytical systems capable of seamlessly integrating textual sentiment, numerical ratings, and multi-level (regional/hotel/aspect) insights are still scarce [11]–[13].

In summary, although ABSA is well established for extracting fine-grained insights from hotel reviews [6], [9], it is often treated as an isolated modeling task. The absence of unified, end-to-end frameworks that integrate textual sentiment

with numerical ratings and enable systematic regional and cross-hotel aggregation continues to limit the translation of these insights into actionable strategies. This persistent gap motivates the development of system-oriented approaches such as *SentimentLens*, which aim to bridge modeling accuracy with practical, data-driven decision making in the hospitality sector.

III. METHODOLOGY

A. Raw Data Sources

Raw hotel reviews are collected using Apify, a web-scraping platform that provides reliable tools for extracting publicly available online content. Reviews are gathered from two major online travel platforms, Google Reviews and TripAdvisor. These platforms were selected due to their widespread use and the availability of both textual feedback and structured rating information. The scraping process extracts only content that users have voluntarily shared publicly on the respective platforms. No private, sensitive, or personally identifiable user data are collected. Each record includes structured attributes such as hotel name, city, province, star rating, and traveler trip type (when available), along with unstructured free-text reviews. The dataset covers hotels from all nine provinces of Sri Lanka, with more than ten hotels per province. For each hotel, the latest 100 reviews are collected, resulting in a corpus of over 10,000 raw hotel reviews. This geographically balanced dataset supports region-level and hotel-level analysis.

B. Data Preprocessing

The collected data are initially stored as raw CSV files for each province. Preprocessing steps include extracting only relevant fields, eliminating empty records, removing duplicate reviews, and merging them into a single unified dataset. Reviews with missing essential metadata or unusable text are discarded. The resulting cleaned dataset is then passed through the text-based and rating-based analytical pipelines.

C. Aspect-Based Sentiment Analysis Framework

The foundational integration layer of the proposed system is a custom ABSA pipeline that integrates Aspect Term Extraction (ATE) and Aspect Sentiment Classification (ASC) into a unified workflow. The entire pipeline is self-hosted and operates locally, without reliance on third-party large language model APIs, enabling scalable, cost-free, and privacy-preserving analysis.

ABSA is performed using *Instruct-DeBERTa* [14], a hybrid instruction-based framework that jointly handles aspect extraction and sentiment classification. The model uses instruction-driven prompts to identify both explicit and implicit aspect terms from review text and simultaneously predicts the corresponding sentiment polarity (positive, negative, or neutral) for each extracted aspect. This unified approach enables coherent modeling of fine-grained opinions by linking aspect detection and sentiment interpretation within a single framework. This enables context-aware sentiment prediction when multiple aspects occur in the same review. To support higher-level analysis, aspect terms are mapped to semantic categories

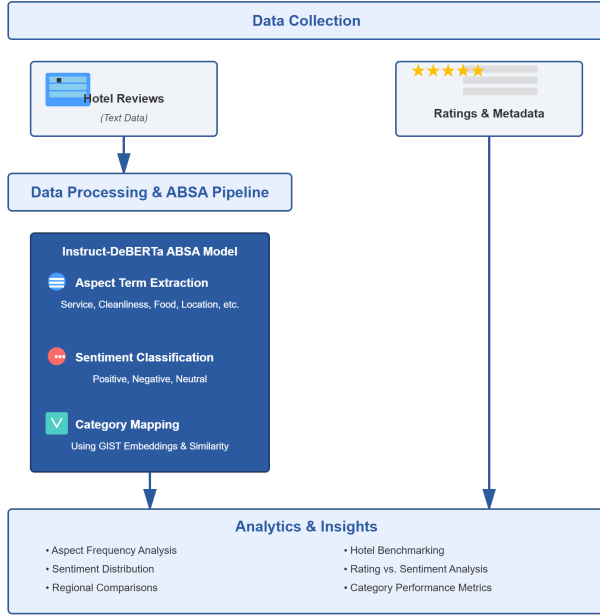


Fig. 1: **System architecture of *SentimentLens*.** The framework ingests raw review text together with structured meta-data, performs aspect extraction and aspect-level sentiment classification, maps extracted aspects into standardized service categories, and then aggregates the outputs into province-level, hotel-level, and cross-modal analytical views for downstream reconciliation.

(Facilities, Food and Dining, Room Quality, Staff, Location, Booking Process) using centroid-based embedding prototypes. Aspect terms and category keywords are embedded using GIST Sentence Transformers [15], and each aspect is assigned to the category with the highest cosine similarity above a threshold of 0.42. This threshold was empirically selected to balance category assignment coverage and semantic precision. The pipeline is applied to over 10,000 reviews, producing structured triples (*aspect: category: sentiment*) stored in CSV format for downstream analysis.

D. Analytical Representation of Aspect-Level Sentiment

Each raw hotel review is decomposed into multiple aspect-level sentiment entries of the form

$$(\text{review}, \text{aspect}, \text{category}, \text{sentiment}).$$

After filtering invalid, missing, or generic categories, each remaining entry represents a single opinion toward a concrete service dimension. This produces an aspect-level dataset $\mathcal{D}$ with one row per aspect mention, enabling aggregation across hotels, provinces, and categories.

Sentiment polarity is mapped to a numerical score

$$s_i = \begin{cases} +1, & \text{positive sentiment,} \\ 0, & \text{neutral sentiment,} \\ -1, & \text{negative sentiment,} \end{cases} \quad (1)$$

where s_i denotes the sentiment score of the i -th aspect mention. Let $\mathcal{A}$ be the set of all aspect mentions, $\mathcal{A}_c$ the subset belonging to category c , $\mathcal{A}_p$ the subset originating from province p , and $\mathcal{A}_{c,p}$ the subset corresponding jointly to category c in province p .

E. Category- and Province-Level Sentiment Aggregation

Category importance is defined as the relative frequency with which a category is mentioned:

$$w_c = \frac{|\mathcal{A}_c|}{\sum_{c'} |\mathcal{A}_{c'}|}, \quad (2)$$

where w_c is the importance weight of category c .

Average sentiment per category is computed as

$$\bar{s}_c = \frac{1}{|\mathcal{A}_c|} \sum_{i \in \mathcal{A}_c} s_i. \quad (3)$$

Overall sentiment per province is computed as

$$\bar{s}_p = \frac{1}{|\mathcal{A}_p|} \sum_{i \in \mathcal{A}_p} s_i. \quad (4)$$

To analyze regional variation across service dimensions, a province–category sentiment matrix is defined as

$$\mathbf{S}_{p,c} = \bar{s}_{c,p}, \quad (5)$$

where $\bar{s}_{c,p}$ is the mean sentiment of category c in province p . For profile-based comparisons, each province is represented by the vector

$$\mathbf{v}_p = [\bar{s}_{p,1}, \bar{s}_{p,2}, \dots, \bar{s}_{p,C}],$$

where C is the number of standardized service categories.

F. Comparative Indices and Stability Measures

For each category c , the best-performing province is defined as

$$p_c^* = \arg \max_p \bar{s}_{c,p}. \quad (6)$$

Inter-provincial variability for category c is measured using the standard deviation

$$\sigma_c = \sqrt{\frac{1}{P} \sum_{p=1}^P (\bar{s}_{c,p} - \mu_c)^2}, \quad (7)$$

where P is the number of provinces and μ_c is the mean sentiment of category c across all provinces.

To integrate category importance with performance, a Province Competitiveness Index (PCI) is defined as

$$\text{PCI}_p = \sum_c w_c \cdot \bar{s}_{c,p}. \quad (8)$$

This index gives greater emphasis to categories that travelers mention more frequently.

G. Co-occurrence, Entropy, and Hotel Archetype Discovery

Let N_{c_1, c_2} denote the number of reviews in which categories c_1 and c_2 co-occur. These values form a category co-occurrence matrix $\mathbf{C}$ that is used to study interactions among service dimensions.

To quantify uncertainty in category-level experience, sentiment entropy is computed as

$$H(c) = - \sum_{k \in \{-1, 0, +1\}} P(s = k | c) \log P(s = k | c), \quad (9)$$

where $P(s = k | c)$ denotes the probability that sentiment class k occurs within category c .

For hotel archetype analysis, each hotel h is represented by a category-level sentiment vector

$$\mathbf{x}_h = [\bar{s}_{h,1}, \dots, \bar{s}_{h,C}].$$

After standardization, K-Means clustering is applied by minimizing

$$\min_{\{\mathcal{C}_k\}} \sum_{k=1}^K \sum_{\mathbf{x}_h \in \mathcal{C}_k} \|\mathbf{x}_h - \boldsymbol{\mu}_k\|^2, \quad (10)$$

where $\mathcal{C}_k$ and $\boldsymbol{\mu}_k$ denote cluster k and its centroid, respectively. The selected number of clusters is determined using the elbow method.

H. Opportunity Gap Formulation

To compare category importance against performance on a common scale, sentiment is normalized to $[0, 1]$ as

$$\tilde{s}_{c,p} = \frac{\bar{s}_{c,p} + 1}{2}. \quad (11)$$

An opportunity score is then defined as

$$\text{Opp}_{c,p} = w_c - \tilde{s}_{c,p}, \quad (12)$$

where high values indicate categories that are heavily discussed but underperform relative to guest expectations.

I. Rating- and Trip-Type-Based Analysis

In parallel with the text-based pipeline, the structured ratings dataset is cleaned by removing invalid or missing trip-type entries. Province-level mean rating is computed as

$$\bar{r}_p = \frac{1}{N_p} \sum_{i=1}^{N_p} r_{i,p}, \quad (13)$$

where N_p is the number of ratings available for province p .

To assess regional differences in ratings, a one-way ANOVA is performed under the null hypothesis

$$H_0 : \bar{r}_1 = \bar{r}_2 = \dots = \bar{r}_P.$$

Pairwise post-hoc comparisons are subsequently examined using Tukey's HSD test. To analyze the association between province and trip type, a chi-square test of independence is used with statistic

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad (14)$$

where O_{ij} and E_{ij} denote observed and expected counts, respectively.

J. Cross-Modal Conflict Identification

To reconcile textual sentiment with numerical ratings, SentimentLens identifies latent conflicts between the normalized rating signal and aspect-level sentiment. A conflict set is defined as

$$C = \{(h, c) \mid |R_{\text{norm}} - S_a| > \tau\} \quad (15)$$

where h denotes a hotel, c a service category, R_{norm} the normalized overall rating, S_a the category-level sentiment score, and τ an empirically selected discrepancy threshold. This formulation captures cases in which high overall ratings mask recurring category-specific weaknesses.

IV. RESULTS

This section presents the empirical findings obtained from 46,565 aspect-level sentiment mentions extracted from 100 hotels distributed across the nine provinces of Sri Lanka.

A. Sentiment-Based Analysis of Traveler Reviews

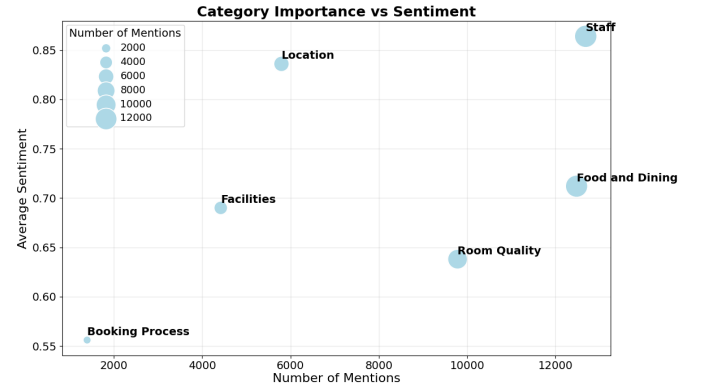


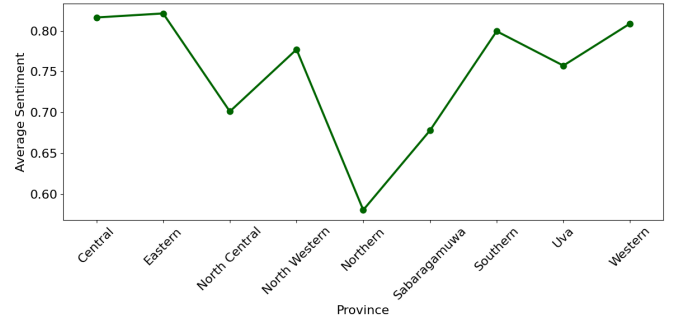
Fig. 2: **Global category importance versus average sentiment.** The figure shows which service categories dominate traveler discourse and how positively they are perceived on average. *Staff* occupies the strongest performance region, while *Booking Process* and *Room Quality* appear as comparatively weaker categories despite their importance, highlighting where operational weaknesses persist.

At the global category level, the guest discourse is dominated by staff, *Food and dining* and *Room Quality* (Fig.2). *Staff* achieves the strongest sentiment performance, suggesting that interpersonal service is a national strength. In contrast, *Booking Process* and *Room Quality* receive weaker sentiment despite being frequently discussed, indicating persistent operational weaknesses in high-visibility service dimensions.

At the province level, Eastern, Central, Western, and Southern Provinces show the strongest overall sentiment (Fig.3b), consistent with more mature tourism ecosystems, while Northern Province records the weakest, reflecting persistent issues in service consistency and accommodation quality. When examined jointly with service category, *Staff* sentiment remains consistently high across most provinces, whereas *Room Quality*, *Facilities*, and *Booking Process* show strong regional



(a) Province-category sentiment heatmap



(b) Average overall rating by province

Fig. 3: Regional sentiment structure across service categories. (a) The heatmap makes cross-province strengths and weaknesses immediately visible and shows that *Staff* remains relatively strong across most regions, whereas *Room Quality*, *Facilities*, and *Booking Process* vary more sharply by province. (b) Average overall ratings by province show that Eastern, Central, and Western Provinces achieve the highest scores, while Northern Province records the lowest, a pattern that closely mirrors the sentiment-based provincial ranking. **Together, these panels confirm that regional performance differences are multidimensional and consistent across both textual and numerical feedback modalities.**

disparities, suggesting that physical infrastructure and operational reliability vary more substantially than staff-related experiences.

The category-wise leader analysis (Fig.3a) shows that no single province dominates every dimension. Central Province performs especially well in *Location*, *Facilities*, and *Booking Process*, while Eastern Province leads in *Staff* and *Food and Dining*. Variation analysis confirms that *Room Quality* and *Facilities* are the least stable categories across provinces, whereas *Staff* and *Location* function as broad national assets. The province competitiveness analysis ranks Eastern Province as the strongest overall performer and Northern Province last, consistent with both category-level and province-level observations.

The co-occurrence analysis reveals that *Staff* is strongly linked with *Food and Dining* and *Room Quality*, suggesting that service experiences are not perceived in isolation. The entropy analysis further shows that *Booking Process* and *Room Quality* exhibit the highest experiential uncertainty, while *Staff* remains the most consistently positive dimension. Hotel-level clustering identifies three archetypes ($K = 3$), premium, mid-tier, and struggling, with low-performing hotels concentrated in Northern and Sabaragamuwa Provinces and stronger hotels in Eastern, Central, Southern, and Western Provinces. Finally, the importance-performance analysis highlights *Room Quality* and *Food and Dining* in Northern, North Central, and Sabaragamuwa Provinces as the clearest high-priority opportunity gaps.

B. Rating- and Trip-Type-Based Analysis

The rating distribution is strongly skewed toward the upper end of the scale, indicating that most hotels receive favorable overall evaluations. However, this apparent positivity does not eliminate meaningful differences across provinces. Southern, Central, and Eastern Provinces achieve the highest average ratings, while Northern Province records the lowest. This

mirrors the province-level sentiment ranking and suggests that the structured rating signal is broadly aligned with the sentiment extracted from text.

Trip-type analysis shows that average ratings vary only slightly across traveler categories, indicating that hotels in the dataset tend to deliver broadly similar overall satisfaction to families, couples, solo travelers, and business travelers. In contrast, the distribution of trip types across provinces is not uniform. Leisure-oriented provinces show stronger concentrations of families and couples, whereas Western Province has a comparatively larger share of business travelers. This indicates that while traveler composition differs geographically, the average rating signal itself remains comparatively stable across traveler types.

The inferential analysis reinforces these descriptive observations. Province-level differences in ratings are statistically significant, confirming that regional quality differences are not due to random fluctuation. Post-hoc comparisons identify Northern Province as significantly lower-rated than most other provinces, while Southern and Uva Provinces appear among the stronger performers. By contrast, ratings do not differ significantly across trip types, supporting the descriptive finding that satisfaction is relatively uniform across traveler groups. The chi-square analysis further confirms that province and trip type are associated, meaning that different regions attract different mixes of travelers even when their average rating levels by traveler type remain similar.

C. Cross-Modal Conflict Identification

To enable systematic reconciliation between textual sentiment and numerical ratings, we identify latent conflicts as discrepancies between the normalized rating signal and aggregated aspect-level sentiment. Intuitively, a conflict arises when overall satisfaction remains high while specific service dimensions exhibit weaker sentiment.

The overall rating is normalized to the $[0, 1]$ range using min-max scaling over the original $[1, 5]$ scale, ensuring direct comparability with sentiment scores. For example, an average rating of 4.24 corresponds to a normalized value of 0.81, as observed in the Northern Province. Conflicts are identified at the province-category level by comparing normalized ratings with category-level sentiment scores, as defined in Equation (15). Larger gaps indicate stronger disagreement between overall satisfaction and fine-grained service experience.

TABLE I: Examples of cross-modal conflicts. Larger gaps indicate stronger disagreement between rating and sentiment.

Province	Category	R_{norm}	S_a	Gap
Northern Province	Room Quality	0.81	0.40	0.41
Northern Province	Facilities	0.81	0.46	0.35
Sabaragamuwa Province	Food	0.85	0.61	0.24

a) *Illustrative examples.*: These examples reveal *latent service gaps*, where relatively strong overall ratings mask weaker performance in specific categories. For instance, in the Northern Province, despite a normalized rating of 0.81, sentiment toward *Room Quality* drops to 0.40, indicating recurring dissatisfaction that is not reflected in the aggregate rating. Similar discrepancies are observed for *Facilities* in the same province and *Food and Dining* in Sabaragamuwa Province. Such patterns highlight a key limitation of rating-only analysis. While ratings capture broad impressions, they often obscure category-specific weaknesses. By contrast, aspect-level sentiment provides fine-grained visibility into operational performance. The proposed cross-modal conflict identification mechanism therefore enables detection of targeted improvement areas that remain hidden when relying on a single modality.

V. DISCUSSION AND CONCLUSION

The *SentimentLens* framework reveals clear geographic disparities in hotel performance across Sri Lankan provinces, highlighting consistent national strengths in *Staff* and *Location* alongside structural weaknesses in *Room Quality*, *Facilities*, and *Booking Process*. This also uncovers latent inconsistencies where strong overall ratings may mask weaker category-level experiences. These findings demonstrate that integrating textual sentiment with numerical ratings provides a more comprehensive and reliable basis for data-driven decision-making in the hospitality sector.

Overall, this study introduced *SentimentLens*, a scalable framework that integrates Aspect-Based Sentiment Analysis (ABSA) with structured rating analytics to generate interpretable, multi-level insights from large-scale hotel reviews. Using over 10,000 publicly available reviews across all nine provinces of Sri Lanka, the framework demonstrates strong capability in identifying actionable service gaps, regional disparities, and hidden performance issues. Although evaluated in a national tourism context, the approach is generalizable, scalable, and adaptable to other regions and multilingual settings.

Future work will focus on extending the framework with real-time data integration, predictive modeling, intelligent recommendation mechanisms, and cross-domain generalization with deeper multimodal fusion to support more robust, proactive, and adaptive decision-making in dynamic environments.

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